Amazon EC2 – Associate

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Private vs Public IP (IPv4)

- Networking has two sorts of IPs. IPv4 and IPv6:
  - IPv4: `1.160.10.240`
  - IPv6: `3ffe:1900:4545:3:200:f8ff:fe21:67cf`

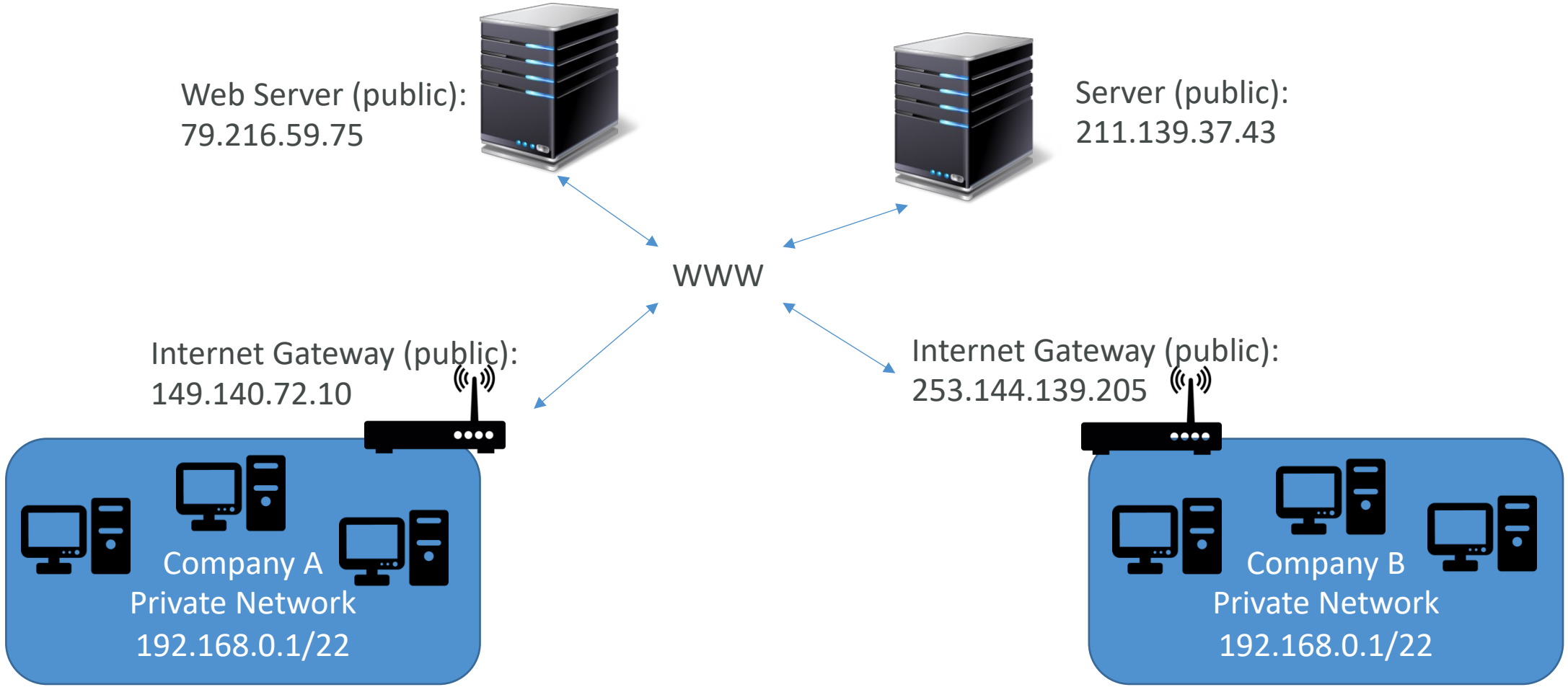
- In this course, we will only be using IPv4.
- IPv4 is still the most common format used online.
- IPv6 is newer and solves problems for the Internet of Things (IoT).

- IPv4 allows for 3.7 billion different addresses in the public space
- IPv4: [0-255].[0-255].[0-255].[0-255].

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Private vs Public IP (IPv4) Example



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Private vs Public IP (IPv4) Fundamental Differences

- Public IP:
  - Public IP means the machine can be identified on the internet (WWW)
  - Must be unique across the whole web (not two machines can have the same public IP).
  - Can be geo-located easily

- Private IP:
  - Private IP means the machine can only be identified on a private network only
  - The IP must be unique across the private network
  - BUT two different private networks (two companies) can have the same IPs.
  - Machines connect to WWW using a NAT + internet gateway (a proxy)
  - Only a specified range of IPs can be used as private IP

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Elastic IPs

- When you stop and then start an EC2 instance, it can change its public IP.
- If you need to have a fixed public IP for your instance, you need an Elastic IP
- An Elastic IP is a public IPv4 IP you own as long as you don't delete it
- You can attach it to one instance at a time

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Elastic IP

- With an Elastic IP address, you can mask the failure of an instance or software by rapidly remapping the address to another instance in your account.

- You can only have 5 Elastic IP in your account (you can ask AWS to increase that).

- Overall, try to avoid using Elastic IP:
  - They often reflect poor architectural decisions
  - Instead, use a random public IP and register a DNS name to it
  - Or, as we'll see later, use a Load Balancer and don't use a public IP

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Private vs Public IP (IPv4) In AWS EC2 – Hands On

- By default, your EC2 machine comes with:
  - A private IP for the internal AWS Network
  - A public IP, for the WWW.

- When we are doing SSH into our EC2 machines:
  - We can't use a private IP, because we are not in the same network
  - We can only use the public IP.

- If your machine is stopped and then started, the public IP can change

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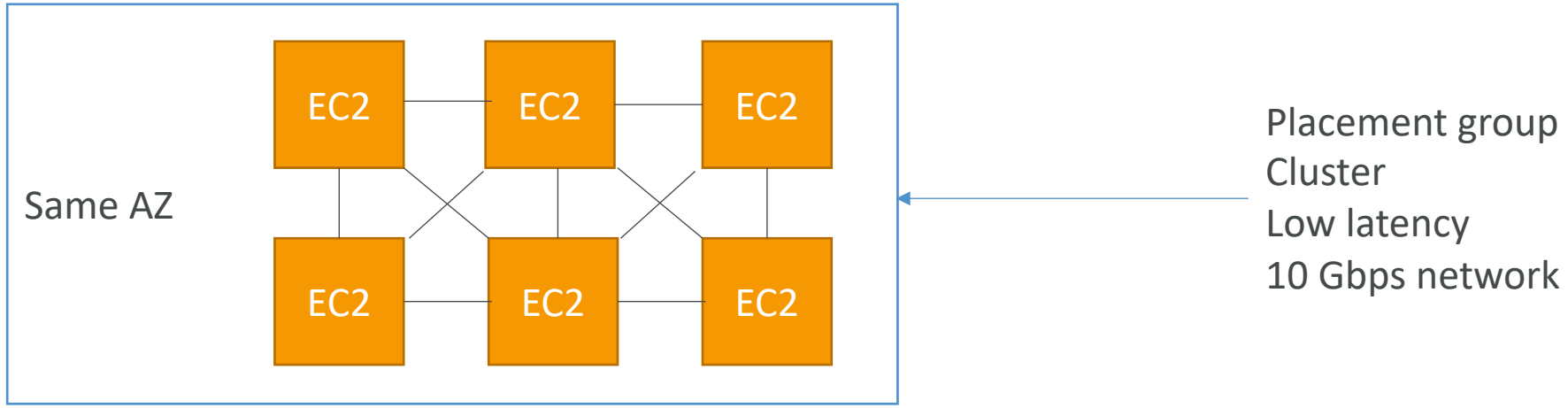
Placement Groups

- Sometimes you want control over the EC2 Instance placement strategy
- That strategy can be defined using placement groups
- When you create a placement group, you specify one of the following strategies for the group:
  - *Cluster*—clusters instances into a low-latency group in a single Availability Zone
  - *Spread*—spreads instances across underlying hardware (max 7 instances per group per AZ)
  - *Partition*—spreads instances across many different partitions (which rely on different sets of racks) within an AZ. Scales to 100s of EC2 instances per group (Hadoop, Cassandra, Kafka)
  - *Precision time*— direct access to high-precision time sources

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Placement Groups
Cluster

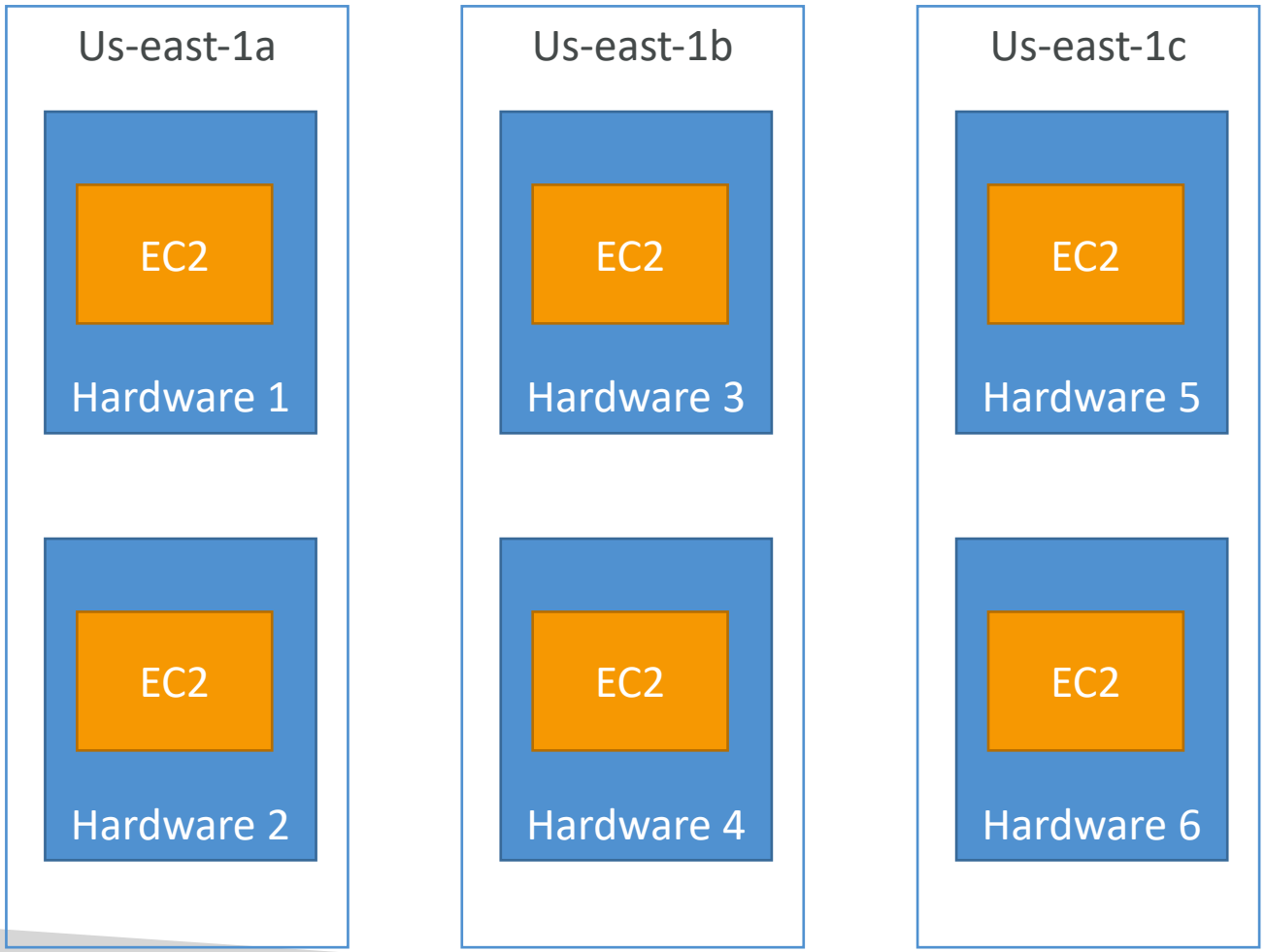


- Pros: Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)
- Cons: If the AZ fails, all instances fails at the same time
- Use case:
  - Big Data job that needs to complete fast
  - Application that needs extremely low latency and high network throughput

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Placement Groups Spread

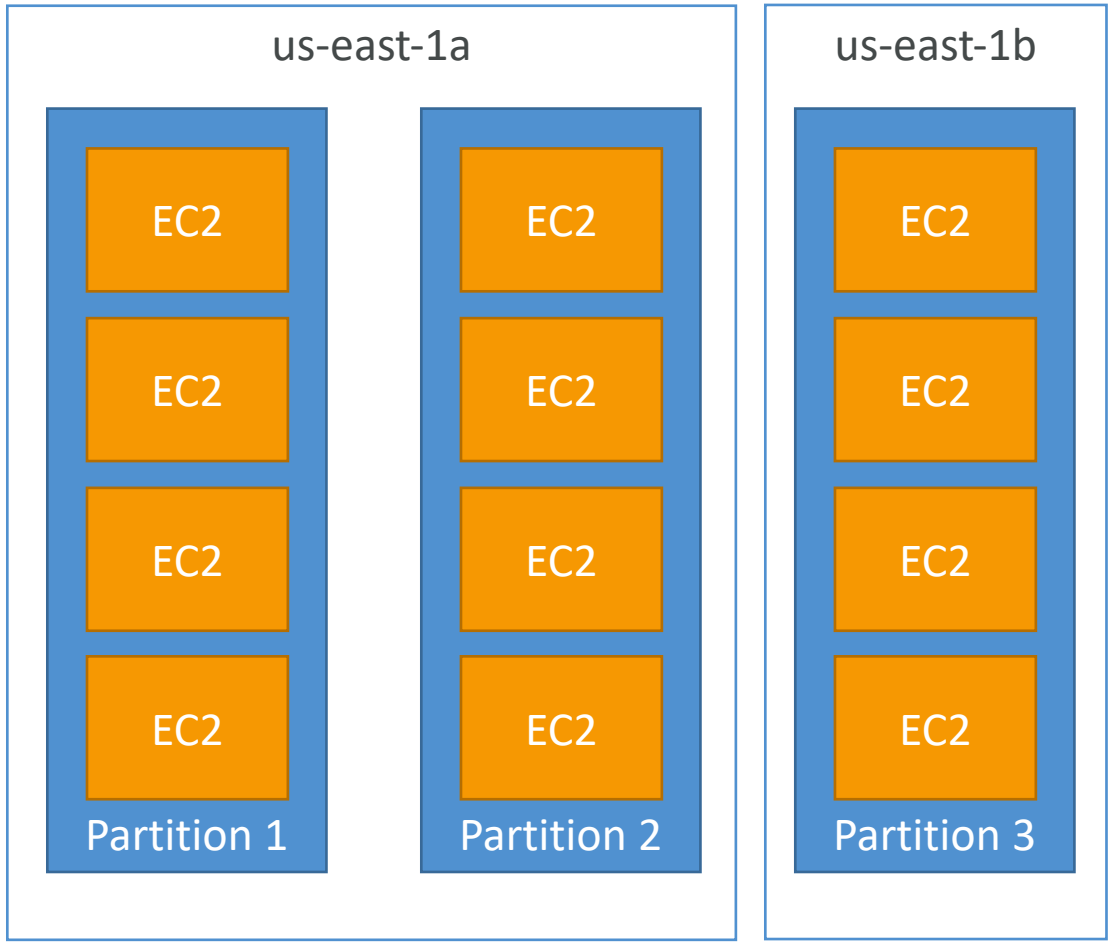


- Pros:
  - Can span across Availability Zones (AZ)
  - Reduced risk is simultaneous failure
  - EC2 Instances are on different physical hardware
- Cons:
  - Limited to 7 instances per AZ per placement group
- Use case:
  - Application that needs to maximize high availability
  - Critical Applications where each instance must be isolated from failure from each other

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Placements Groups Partition

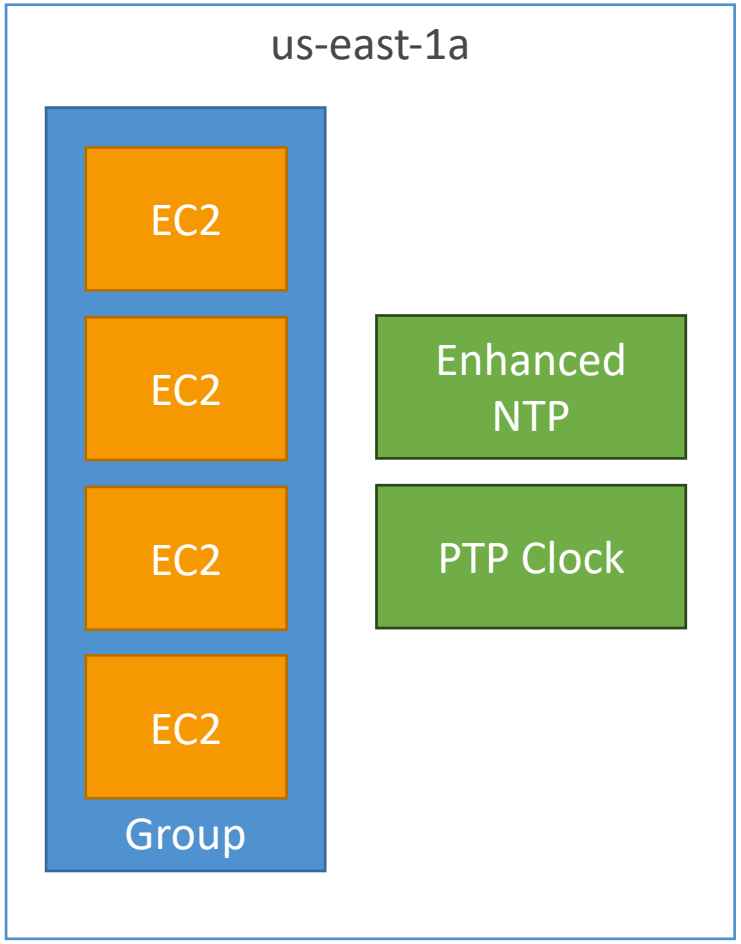


- Up to 7 partitions per AZ
- Can span across multiple AZs in the same region
- Up to 100s of EC2 instances
- The instances in a partition do not share racks with the instances in the other partitions
- A partition failure can affect many EC2 but won't affect other partitions
- EC2 instances get access to the partition information as metadata
- Use cases: HDFS, HBase, Cassandra, Kafka

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Placements Groups Precision time



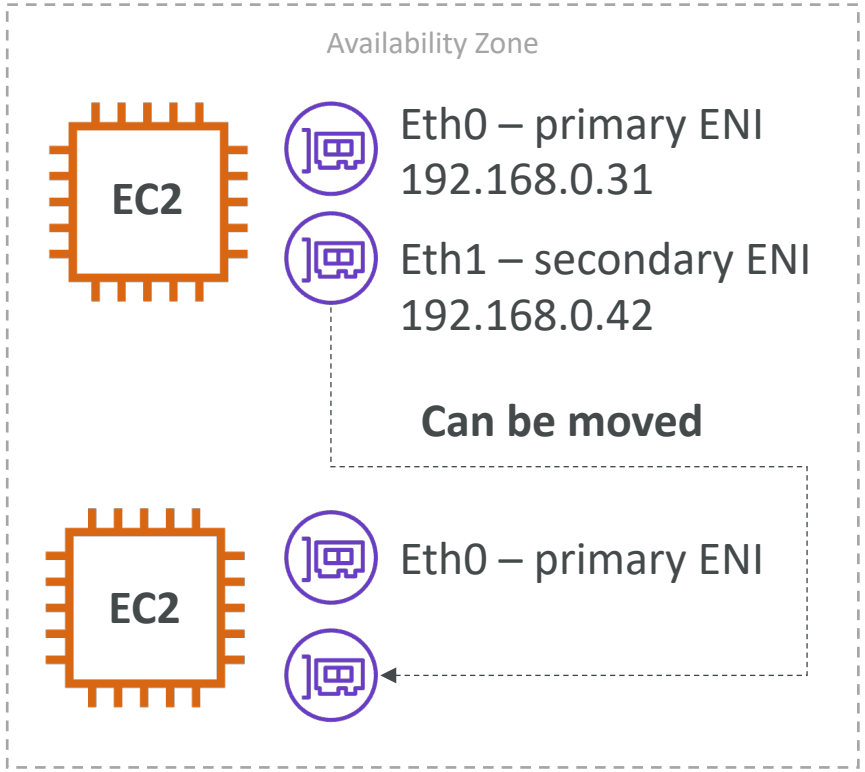
- Instances placed on hardware with direct access to high-precision time sources
- Enhanced Amazon Time Sync Service (more accurate than standard NTP)
- PTP Hardware Clock + hardware packet timestamping (Linux only)
- Use cases: distributed databases, financial timestamping, event ordering

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Elastic Network Interfaces (ENI)

- Logical component in a VPC that represents a **virtual network card**
- The ENI can have the following attributes:
  - Primary private IPv4, one or more secondary IPv4
  - One Elastic IP (IPv4) per private IPv4
  - One Public IPv4
  - One or more security groups
  - A MAC address
- You can create ENI independently and attach them on the fly (move them) on EC2 instances for failover
- Bound to a specific availability zone (AZ)



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EC2 Hibernate

- We know we can stop, terminate instances
  - **Stop** – the data on disk (EBS) is kept intact in the next start
  - **Terminate** – any EBS volumes (root) also set-up to be destroyed is lost

- On start, the following happens:
  - First start: the OS boots & the EC2 User Data script is run
  - Following starts: the OS boots up
  - Then your application starts, caches get warmed up, and that can take time!

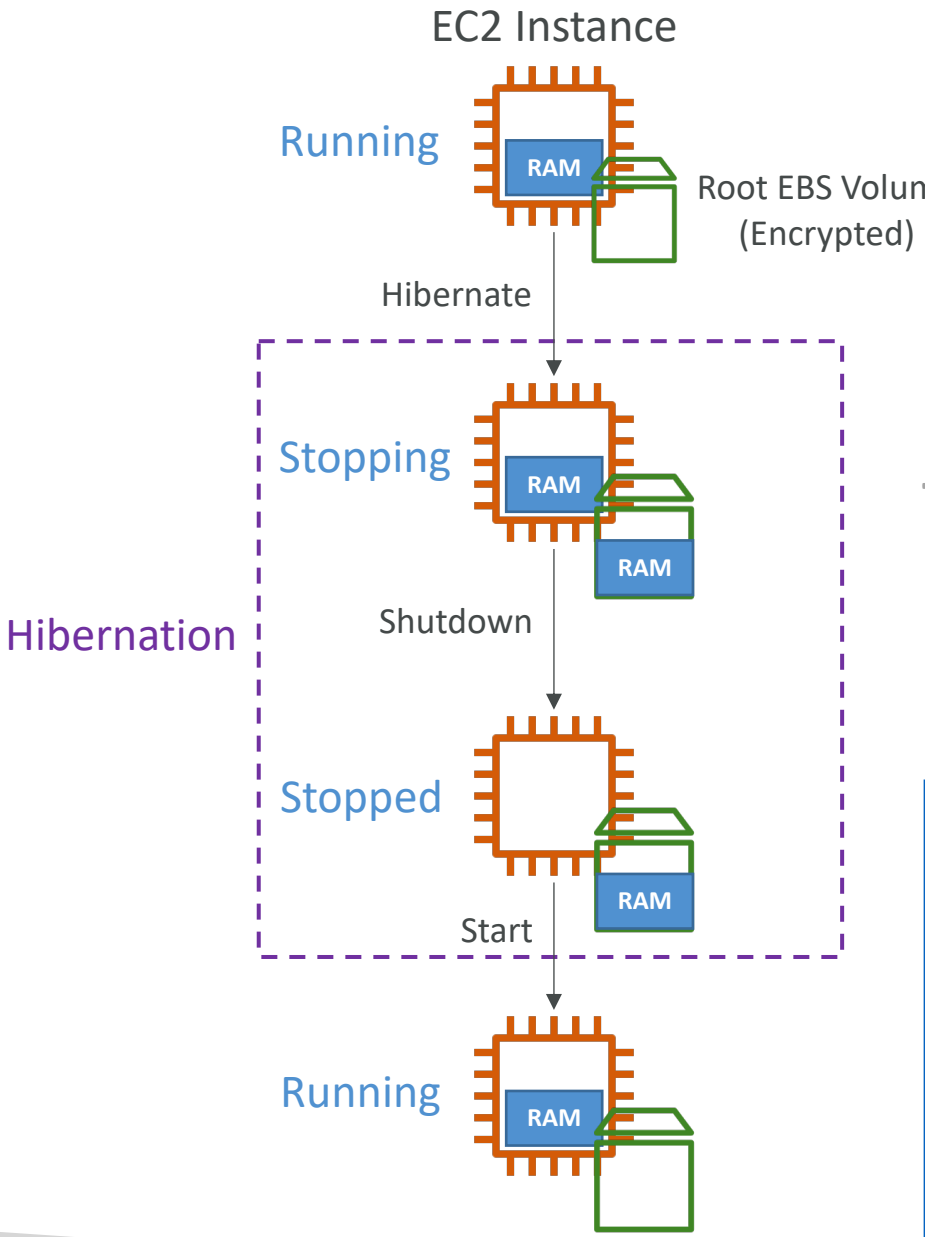
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EC2 Hibernate

- Introducing **EC2 Hibernate**:
  - The in-memory (RAM) state is preserved
  - The instance boot is much faster! (the OS is not stopped / restarted)
  - Under the hood: the RAM state is written to a file in the root EBS volume
  - The root EBS volume must be encrypted

- Use cases:
  - Long-running processing
  - Saving the RAM state
  - Services that take time to initialize



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EC2 Hibernate – Good to know

- **Supported Instance Families** – C3, C4, C5, I3, M3, M4, R3, R4, T2, T3, …
- **Instance RAM Size** – must be less than 150 GB.
- **Instance Size** – not supported for bare metal instances.
- **AMI** – Amazon Linux 2, Linux AMI, Ubuntu, RHEL, CentOS & Windows…
- **Root Volume** – must be EBS, encrypted, not instance store, and large
- Available for **On-Demand**, **Reserved** and **Spot** Instances

- An instance can **NOT** be hibernated more than 60 days

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